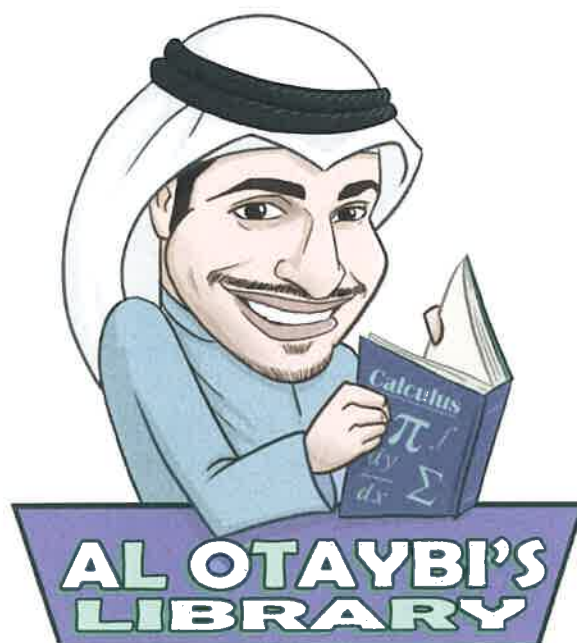


Kuwait University

2021/2022

Fall course



Calculus A

Final

Sections : 5.3 – 6.1

5.3 The Fundamental Theorem of Calculus.

6.1 Areas Between Curves.

(20)

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The Fundamental Theorem

$$\frac{d}{dx} \int_{h(x)}^{g(x)} f(t) dt = f(g(x))g'(x) - f(h(x))h'(x)$$

نظرية تستخدم لاشتقاق التكاملات المحددة

عوض

اشتق

-

عوض

اشتق

Examples

$$\frac{d}{dx} \int_{x+1}^{x^3} t^2 dt$$

$$= (x^3)^2 \cdot 3x^2 - (x+1)^2 \cdot (1)$$

$$= (2x^2+1) \cdot (1) - 0$$

ونشفة الرقم رافعا ٠

$$\frac{d}{dx} \int_6^{9x^2+1} 5t^3 dt$$

$$= 5(9x^2+1)^3 \cdot 18x - 0$$

$$\frac{d}{dx} \int_1^3 3t^2 dt$$

$$= 0 - 0 = 0$$

$$\frac{d}{dx} \int_1^x (2t^2 + 1) dt$$

Find $\frac{dy}{dx}$ where

$$y = \int_1^{\cos x} \frac{1-u^2}{1+u^6} du$$

$$\begin{aligned} y' &= \frac{d}{dx} \int_1^{\cos x} \frac{1-u^2}{1+u^6} du \\ &= \frac{1-(\cos x)^2}{1+(\cos x)^6} (-\sin x) - 0 \end{aligned}$$

Find $f'(x)$ where

$$y = \int_{3x}^{x^2} \sqrt{\sin^2 t + 7t^6} dt$$

$$y' = \frac{d}{dx} \int_{3x}^{x^2} \sqrt{\sin^2 t + 7t^6} dt$$

$$= \sqrt{\sin^2(x^2) + 7(x^2)^6} (2x)$$

$$- \sqrt{\sin^2(3x) + 7(3x)^6} (3)$$

Find $f'(x)$ where

$$f(x) = \int_{x^3}^{\tan x} 3t^3 dt$$

$$f'(x) = 3(\tan x)^3 \sec^2 x - 3(x^3)^3 3x^2$$

Find $f'(x)$ where

$$f(x) = \int_{x^3+x}^2 \sqrt{t^2+1} dt$$

$$f'(x) = 0 - \sqrt{(x^3+x)^2+1} (3x^2+1)$$

Find $f'(x)$ where

$$f(x) = \int_{\sin^{-1}x}^{\ln x} \frac{t-1}{t^2+1} dt$$

$$f'(x) = \frac{\ln x - 1}{(\ln x)^2 + 1} \frac{1}{x} - \frac{\sin^{-1}x - 1}{(\sin^{-1}x)^2 + 1} \frac{1}{\sqrt{1-x^2}}$$

Find the derivative of

$$f(x) = \int_{\tan^{-1}x}^{\sqrt{x}} \frac{1}{t^4 + 2} dt - \int_{\sin x}^5 t dt + \int_2^5 \sin t dt$$

$$f'(x) = \frac{1}{(\sqrt{x})^4 + 2} \cdot \frac{1}{2\sqrt{x}} - \frac{1}{(\tan^{-1}x)^4 + 2} \cdot \frac{1}{x^2 + 1} - [0 - \sin x \cos x] + 0 - 0$$

Find the derivative of

$$f(x) = x^2 \int_{\cos x}^{\sinh x} \sqrt{t-1} dt$$

$$f'(x) = 2x \int_{\cos x}^{\sinh x} \sqrt{t-1} dt + x^2 [\sqrt{\sinh x - 1} \cosh x - \sqrt{\cos x - 1} (-\sin x)]$$

Find the derivative of

$$f(x) = \frac{\int_3^{x^2} \sqrt{t-1} dt}{x+1}$$

$$f'(x) = \frac{(x+1) [\sqrt{x^2-1} \cdot 2x - 0] - \int_3^{x^2} \sqrt{t-1} dt (1)}{(x+1)^2}$$



Remark



$$f(x)$$

Domain

Horizontal Asymptotes

Vertical Asymptotes

$$f'(x)$$

Interval of increasing and decreasing

Local Extrema

(Local maximum and Local minimum)

$$f''(x)$$

Interval of concavity

(Concave up – Concave down)

Inflection points

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LIBRARY

Show that $f(x)$ is increasing on $\mathbb{R}$

$$f(x) = 2x + \int_0^{x^3} \sqrt{t^2 + 1} dt + \int_0^1 \sqrt{s^2 + 1} ds$$

$$\begin{aligned} f'(x) &= 2 + \sqrt{(x^3)^2 + 1} \cdot 3x^2 - 0 + 0 - 0 \\ &= 2 + \sqrt{x^6 + 1} \cdot (3x^2) > 0 \quad \forall x \in \mathbb{R} \\ \therefore f'(x) > 0 &\implies \therefore f(x) \text{ is increasing} \end{aligned}$$

Show that $f(x)$ is decreasing on $\mathbb{R}$

$$f(x) = \int_x^0 (\cos t + 11)^{26} dt$$

$$\begin{aligned} f'(x) &= 0 - (\cos x + 11)^{26} (1) \\ &= -(\cos x + 11)^{26} < 0 \quad \forall x \in \mathbb{R} \\ \therefore f'(x) < 0 &\implies \therefore f(x) \text{ is decreasing} \end{aligned}$$

Show that $f(x)$ is constant function on $[1, \infty)$

$$f(x) = 2 \int_1^{3x} \frac{1}{t} dt - 2 \int_2^{3x} \frac{1}{t} dt$$

$$\begin{aligned} f'(x) &= 2 \left[\frac{1}{3x} - 0 \right] - 2 \left[\frac{1}{3x} - 0 \right] \\ &= \frac{2}{x} - \frac{2}{x} = 0 \\ \therefore f'(x) = 0 &\implies \therefore f(x) \text{ is constant} \end{aligned}$$

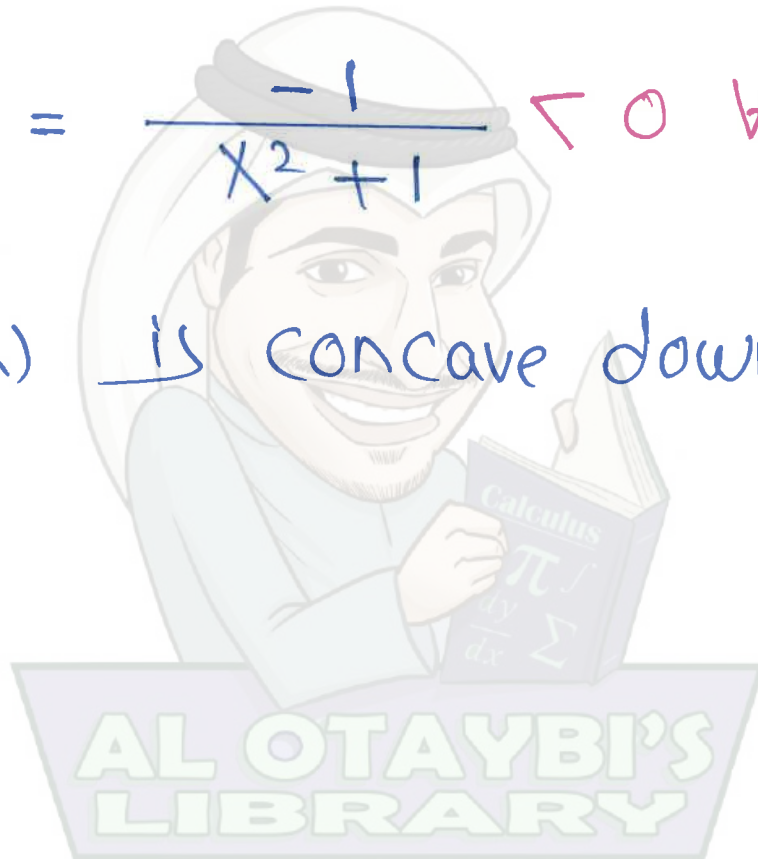
$$\text{Let } f(x) = \int_0^x \cot^{-1} s \, ds \quad x \in \mathbb{R}$$

Show that $f(x)$ is concave down for all x

$$\begin{aligned} f'(x) &= \cot^{-1} x \quad (1) - 0 \\ &= \cot^{-1} x \end{aligned}$$

$$f''(x) = \frac{-1}{x^2 + 1} < 0 \quad \forall x \in \mathbb{R}$$

$\therefore f(x)$ is concave downward.



$$\text{Let } f(x) = \int_0^{x^2} \frac{t-4}{t^2+1} dt$$

Find the intervals on which f is increasing and the intervals on which f is decreasing

Domain of $f(x) = \mathbb{R}$

$$f'(x) = \frac{x^2 - 4}{(x^2)^2 + 1} \quad 2x - 0$$

$$= \frac{2x(x^2 - 4)}{x^4 + 1}$$

$$= 0$$

$$2x(x^2 - 4) = 0$$

$$2x(x-2)(x+2) = 0$$

$$2x = 0 \quad x-2 = 0 \quad x+2 = 0$$

$$x = 0$$

$\in \text{Domain}$

$$x = 2$$

$\in \text{Domain}$

$$x = -2$$

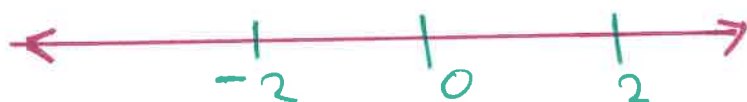
$\in \text{Domain}$

DNE

$$x^4 + 1 = 0$$

$$x^4 = -1$$

None



	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
$2x(x^2 - 4)$	-	+	-	+
$x^4 + 1$	+	+	+	+

$\ominus$

$\oplus$

$\ominus$

$\oplus$

$\therefore f(x)$ is increasing on $(-2, 0)$ and $(2, \infty)$

$\therefore f(x)$ is decreasing on $(-\infty, -2)$ and $(0, 2)$



Extra

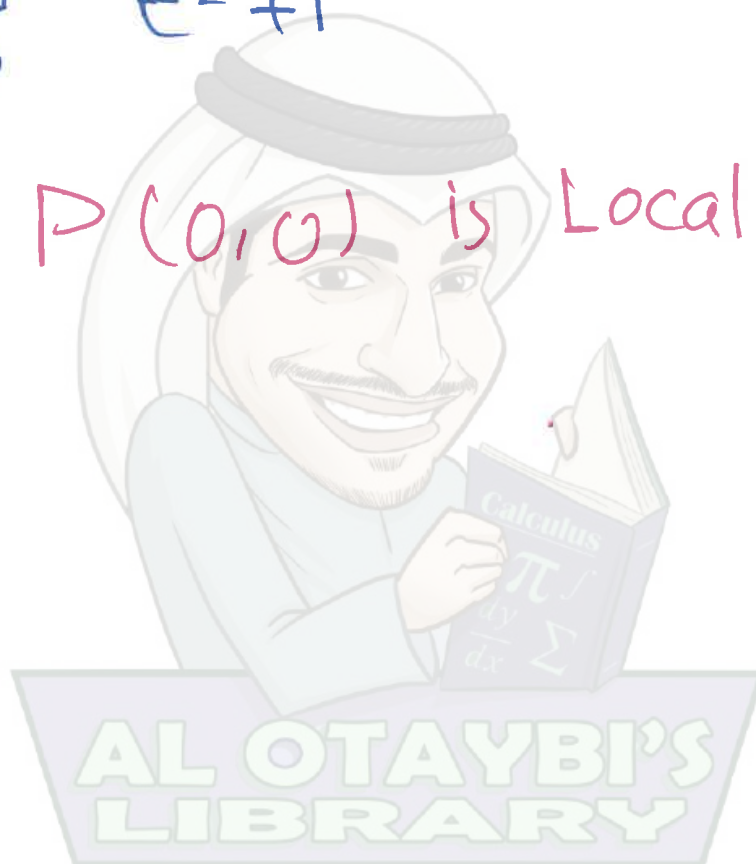
Find the Local maximum point



at $x=0 \longrightarrow$ there is Local maximum

$$y = \int_0^0 \frac{t-4}{t^2+1} dt = 0$$

$\Rightarrow P(0,0)$ is Local maximum point



Find the local maximum point of

$$f(x) = \int_1^{2x-x^2} \frac{1}{t^4 + 2} dt$$

Domain of $f(x) = \mathbb{R}$

$$f'(x) = \frac{1}{(2x-x^2)^4 + 2} (2-2x) - 0$$

$$= \frac{2-2x}{(2x-x^2)^4 + 2}$$

$$= 0$$

$$2-2x=0$$

$$2=2x$$

$$x=1$$

DNE

$$(2x-x^2)^4 + 2 = 0$$

$$(2x-x^2)^4 = -2$$

None



	$(-\infty, 1)$	$(1, \infty)$
$\frac{2-2x}{(2x-x^2)^4 + 2}$	+	-
	+	+

+

-

at $x=1 \rightarrow$ there is Local maximum

$$y = \int_1^{2-1} \frac{1}{t^4 + 2} dt$$

$$= \int_1^1 \frac{1}{t^4 + 2} dt = 0$$

$\Rightarrow (1, 0)$ is the Local maximum

Find the x-coordinates of the points of inflection of the continuous function

$$f(x) = \int_{-5}^{x^2} \frac{1}{3+t^2} dt + \int_1^7 \sqrt{3-t^2} dt$$

Domain of $f(x) = \mathbb{R}$

$$f'(x) = \frac{1}{3+(x^2)^2} \cdot 2x - 0 + 0 - 0$$

$$= \frac{2x}{3+x^4}$$

$$f''(x) = \frac{(3+x^4) \cdot 2 - 2x \cdot 4x^3}{(3+x^4)^2}$$

$$= \frac{6+2x^4-8x^4}{(3+x^4)^2}$$

$$= \frac{6-6x^4}{(3+x^4)^2}$$

$$= 0$$

$$6-6x^4=0$$

$$6=6x^4$$

$$x^4=1 \rightarrow x=1 \in \text{Domain}$$

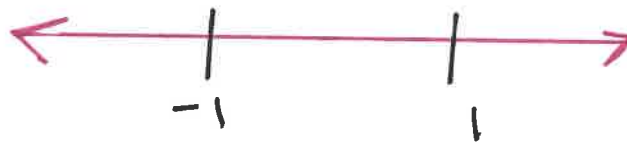
$$x^4=1 \rightarrow x=-1 \in \text{Domain}$$

DNE

$$3+x^4=0$$

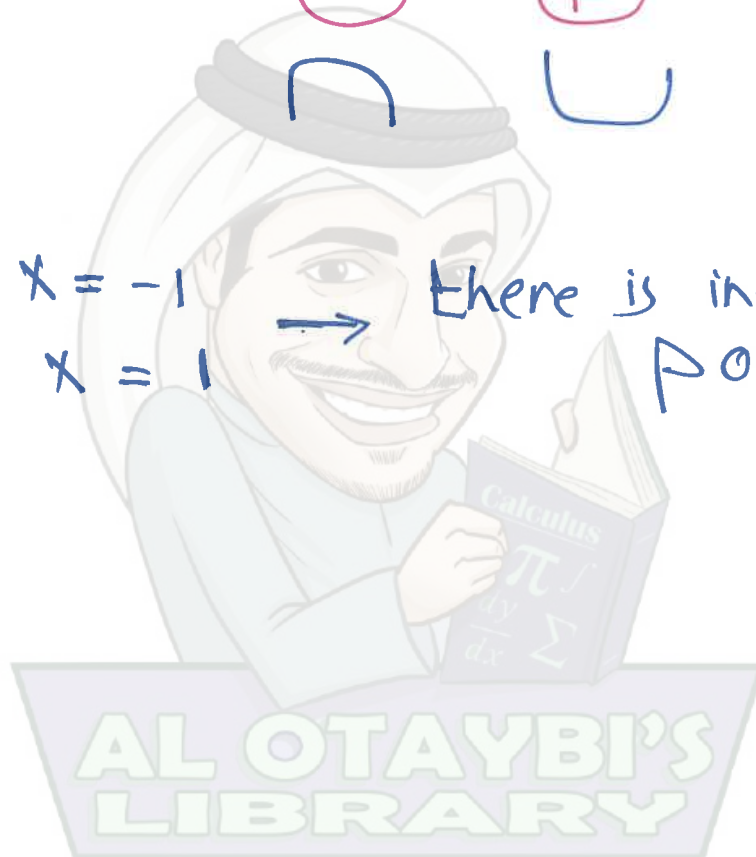
$$x^4=-3$$

None



	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
$\frac{6-6x^4}{(3+x^4)^2}$	$-$	$+$	$-$
	$+$	$+$	$+$
	$\ominus$	$\oplus$	$\ominus$
	$\cap$	$\cup$	$\cap$

$\therefore$ at $x = -1$ $\rightarrow$ there is inflection
 at $x = 1$ points



On what interval is the curve

Jan 2017

$$y = \int_0^x \frac{1-t}{t^2+8} dt \text{ concave downward?}$$

Domain of $f(x)$: $\mathbb{R}$

$$y' = \frac{1-x}{x^2+8} \quad (1) - 0$$

$$= \frac{1-x}{x^2+8}$$

$$y'' = \frac{(x^2+8)(-1) - (1-x)(2x)}{(x^2+8)^2}$$

$$= \frac{-x^2-8-2x+2x^2}{(x^2+8)^2}$$

$$= \frac{x^2-2x-8}{(x^2+8)^2}$$

$$= 0$$

$$x^2-2x-8=0$$

$$(x+2)(x-4)=0$$

$$x=-2 \in \text{Domain}$$

$$x=4 \in \text{Domain}$$

undefind

$$x^2+8=0$$

$$x^2=-8$$

None



y''	$(-\infty, -2)$	$(-2, 4)$	$(4, \infty)$
$\frac{(x+2)(x-4)}{(x^2+8)^2}$	+	-	+

$\therefore f(x)$ is concave downward on $(-2, 4)$

Find the inflection points , if any , of the curve

$$y = \int_0^x (1-t)e^t dt$$

Dec 2017

Domain of $y : \mathbb{R}$

$$y' = (1-x)e^x \quad (1) - 0 \\ = (1-x)e^x$$

$$y'' = \frac{d}{dx} (1-x)e^x = (-1)e^x + (1-x)e^x \\ = e^x [-1 + 1 - x] \\ = e^x (-x) \\ = -xe^x$$

$$= 0$$

$$-xe^x = 0$$

$$-x = 0$$

$$\div (-1)$$

$$x = 0 \\ \in \text{Domain}$$

undefind

None

$$e^x = 0 \\ \text{None}$$

$\therefore f(x)$ is concave up on $(-\infty, 0)$

$\therefore f(x)$ is concave down on $(0, \infty)$

at $x=0 \rightarrow$ inflection point

$$y = \int_0^0 (1-t)e^t dt = 0$$

$\Rightarrow \therefore (0,0)$ is the inflection point

y''	$(-\infty, 0)$	$(0, \infty)$
$-xe^x$	+	-
	$\oplus$	$\ominus$
	$\cup$	$\cap$

$$\text{Let } f(x) = \int_0^{x^2} \frac{1}{\cos t - 3} dt \quad x \in \mathbb{R}$$

Find the local extrema point of $f(x)$

Domain of $f(x) = \mathbb{R}$

$$f'(x) = \frac{1}{\cos(x^2) - 3} \cdot 2x - 0$$

$$= \frac{2x}{\cos(x^2) - 3}$$

$$= 0$$

$$2x = 0$$

$$\div 2$$

$$x = 0$$

$$\in \mathbb{R}$$

DNE

$$\cos(x^2) - 3 = 0$$

$$\cos x^2 = 3$$

None

$$(\because -1 \leq \cos x^2 \leq 1)$$

$$\leftarrow \frac{1}{0} \rightarrow$$

$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{2x}{\cos(x^2) - 3}$	-	+
	-	-
	+	-

at $x=0 \rightarrow$ there is Local max

$$y = \int_0^0 \frac{1}{\cos t - 3} dt = 0$$

$\Rightarrow (0, 0)$ is Local maximum

No Local Minimum.

Let $f(x) = \int_x^{x+3} t(5-t) dx \quad x \in \mathbb{R}$

Show that $f(x)$ has maximum value at $x = 1$

Domain of $f(x) = \mathbb{R}$

$$\begin{aligned} f'(x) &= (x+3) [5 - (x+3)] (1) - x(5-x) (1) \\ &= (x+3) [5 - x - 3] - 5x + x^2 \\ &= (x+3) [2 - x] - 5x + x^2 \\ &= 2x - x^2 + 6 - 3x - 5x + x^2 \\ &= -6x + 6 \end{aligned}$$

$= 0$

$-6x + 6 = 0$

$6x = 6$

$\div 6$

$x = 1 \in \text{Domain}$

DNE

Done

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Sol #1

	$(-\infty, 1)$	$(1, \infty)$
$-6x + 6$	+	-



$\therefore$ at $x=1 \rightarrow$ there is local maximum

Sol #2

$f''(x) = -6$

at $x=1$

$f''(1) = -6 < 0$

$\therefore$ at $x=1 \rightarrow f$ is max.

$$\text{Let } f(x) = \int_1^{x^2+1} \frac{t^2}{\sqrt{t^2+1}} dt$$

Aug 2017

Find the absolute minimum value of f , if any.

Domain of $f(x) = \mathbb{R}$

$$f'(x) = \frac{(x^2+1)^2}{\sqrt{(x^2+1)^2+1}} (2x) - 0$$

$$= \frac{2x (x^2+1)^2}{\sqrt{(x^2+1)^2+1}}$$

$= 0$

$$2x (x^2+1)^2 = 0$$

undefined

$$(x^2+1)^2+1 = 0$$

$$(x^2+1)^2 = -1$$

None

$$2x = 0$$

$\div 2$

$$x = 0$$

$\in$
Domain

$$(x^2+1)^2 = 0$$

$$x^2+1 = 0$$

$$x^2 = -1$$

None

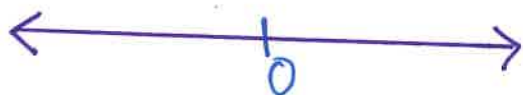
$\therefore f(x)$ is increasing on $(0, \infty)$

$f(x)$ is decreasing on $(-\infty, 0)$

at $x=0 \rightarrow$ L.min

$$y = \int_1^0 \frac{t^2}{\sqrt{t^2+1}} dt = 0$$

$\therefore y(0) = 0$ is the minimum value.



$f'(x)$	$(-\infty, 0)$	$(0, \infty)$
$\frac{2x(x^2+1)^2}{\sqrt{(x^2+1)^2+1}}$	-	+
	+	+

$\ominus$

$\oplus$

$$\text{if } f(x) = \int_0^{3x-x^2} \frac{1}{t^4+4} dt$$

at $x = 3$ Find :

a) The equation of tangent line.

b) The equation of normal line.

$$\begin{aligned} \text{at } x=3 &\longrightarrow y = \int_0^{9-9} \frac{1}{t^4+4} dt \\ &= \int_0^0 \frac{1}{t^4+4} dt = 0 \end{aligned}$$

$$\Rightarrow P(3, 0)$$

$$\begin{aligned} f'(x) &= \frac{1}{(3x-x^2)^4+4} (3-2x) - 0 \\ &= \frac{3-2x}{(3x-x^2)^4+4} \end{aligned}$$

at (3,0)

$$f'(3) = \frac{3-6}{(9-9)^4+4} = \frac{-3}{0+4} = -\frac{3}{4}$$

$$\therefore \begin{aligned} \text{(1) the equation of the tangent line: } &y - y_0 = m(x - x_0) \\ &y - 0 = -\frac{3}{4}(x - 3) \end{aligned}$$

$$\text{(2) the equation of the normal line: } y - 0 = \frac{4}{3}(x - 3)$$

Find an equation of the tangent line to the graph of the function

$$f(x) = \int_{-1}^{x^3} \frac{\ln(t+2)}{t^2+1} dt, \text{ at } x = -1$$

May 2017

at $x = -1 \longrightarrow y = \int_{-1}^{-1} \frac{\ln(t+2)}{t^2+1} dt = 0 \longrightarrow P(-1, 0)$

$$f'(x) = \frac{\ln(x^3+2)}{(x^3)^2+1} (3x^2) - 0$$

$$= \frac{3x^2 \cdot \ln(x^3+2)}{x^6+1}$$

at $P(-1, 0)$

$$f'(-1) = \frac{3(1) \ln(-1+2)}{1+1} = \frac{3 \ln(1)}{2} = \frac{0}{2} = 0$$

$\therefore$ The equation of the tangent line:

$$y - y_0 = m(x - x_0)$$

$$y - 0 = 0(x - (-1))$$

$$y = 0$$



6.1 Area between curves



خطوات الحل



$$x = x$$

$$y = y$$

(1) إيجاد نقاط التقاطع

(2) رسم الدوال



or



المناسب

element

(3) اختيار

(4) تطبيق القانون

$$A = \int_a^b (\text{الصغيرة} - \text{الكبيرة}) \begin{matrix} dx \\ \text{or} \\ dy \end{matrix}$$



element



dx

x

حدود التكامل هي حدود الـ x الدالة الكبيرة هي الأعلى والصغيرة هي الأسفل

dy

y

حدود التكامل هي حدود الـ y الدالة الكبيرة هي اليمين والصغيرة هي اليسار

Set up an integral to find the area of the region bounded by

$$x = y^2 \text{ and } x - y = 2$$

$$x = y^2$$

$$x - y = 2$$

$$x = 2 + y$$

$$x = x$$

$$y^2 = 2 + y$$

$$y^2 - y - 2 = 0$$

$$(y + 1)(y - 2) = 0$$

$$y + 1 = 0$$

$$y - 2 = 0$$

$$y = -1$$

$$y = 2$$

$$x = y^2$$

$$x = 2 + y$$

x	1	0	4
y	-1	0	2

x	1	2	4
y	-1	0	2

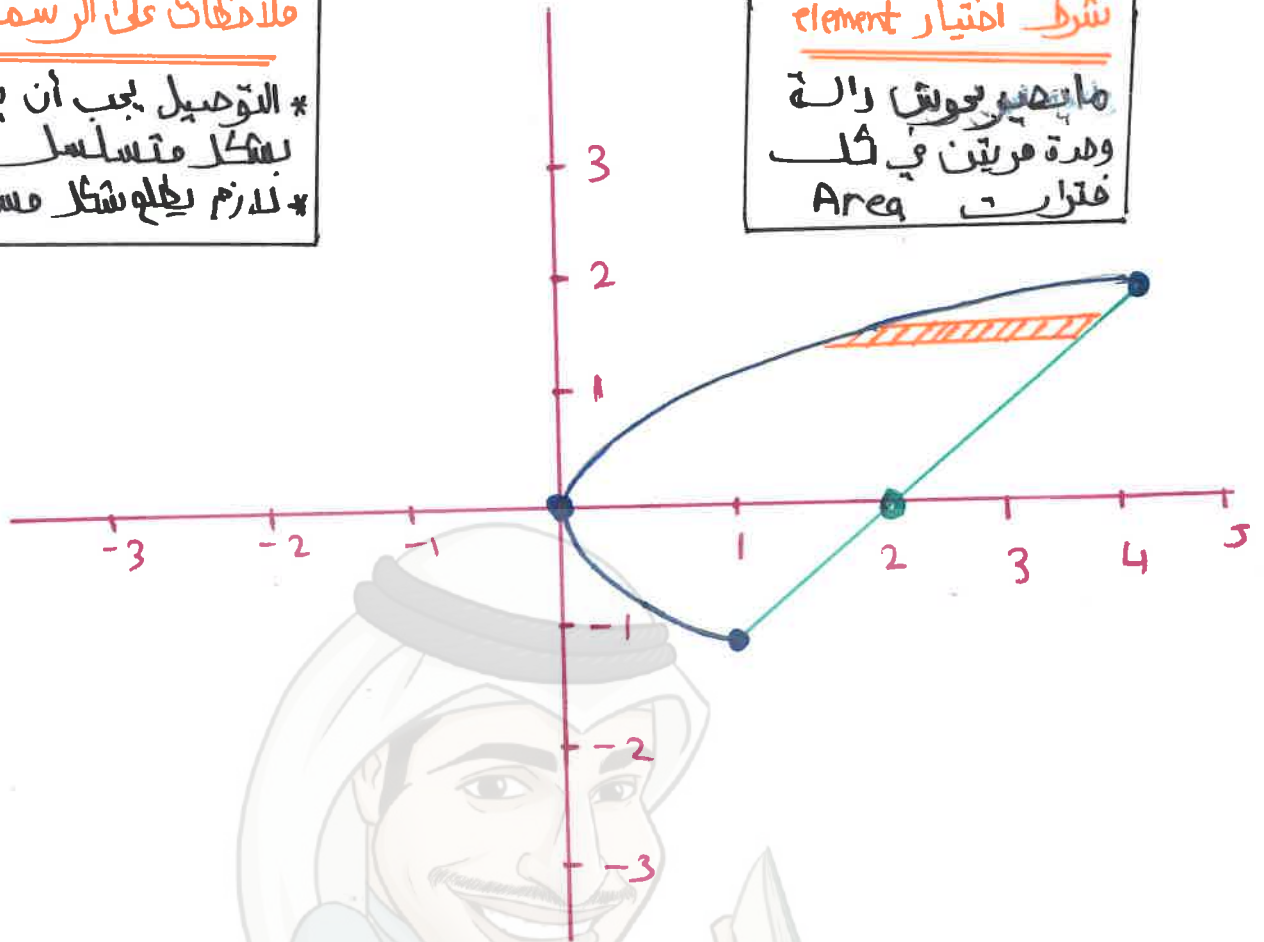
كل نقطتين تقاطع
لازم أن تكون نقطة بينهما

ملاحظات على الرسمة

* التوضيل يجب أن يتم
بشكل متسلسل
* نلاحظ شكل مسك

شرط اختيار element

ما يصير عوض دالة
وهدة مرتين في تلك
فترات Area



$$A = \int_{-1}^4 [2 + y - y^2] dy$$

* قاله داعب أهل التكامل *

Set up

لان قاله

Find the area of the region bounded by the curves

$$x = y^2 \quad \text{and} \quad x = -2y^2 + 3$$

$$x = y^2$$

$$x = -2y^2 + 3$$

$$x = x$$

$$y^2 = -2y^2 + 3$$

$$3y^2 = 3$$

$\div 3$

$$y^2 = 1$$

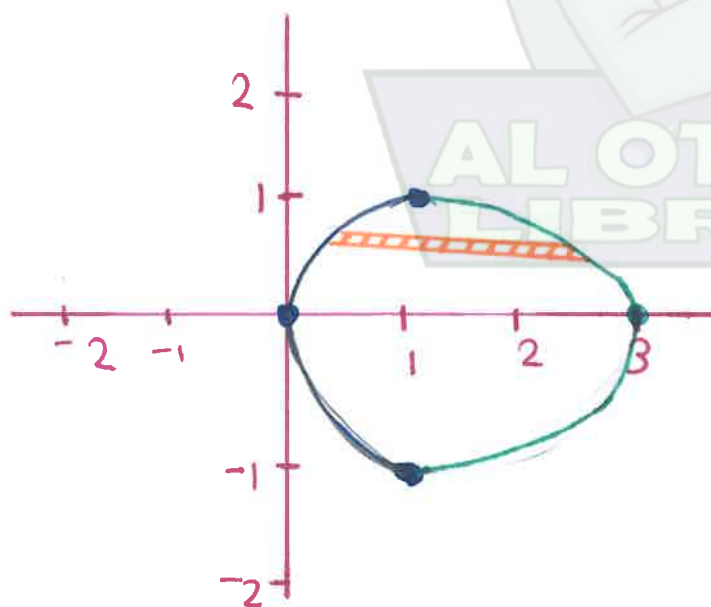
$$y = -1 \quad y = 1$$

$$x = y^2$$

x	1	0	1
y	-1	0	1

$$x = -2y^2 + 3$$

x	1	3	1
y	-1	0	1



$$A = \int_{-1}^1 [-2y^2 + 3 - y^2] dy$$

$$= \int_{-1}^1 (-3y^2 + 3) dy$$

$$= \left[-\frac{3y^3}{3} + 3y \right]_{-1}^1$$

$$= [-1 + 3] - [1 - 3]$$

$$= 2 - [-2]$$

$$= 2 + 2 = 4$$

Let R be the region enclosed by the curves $y = x - 2$ and $y = x^2 - 4$

a) Sketch the region of R .

b) Find the area of the region R .

Dec 2017

$$y = x - 2 \quad y = x^2 - 4$$

$$y = y$$

$$x - 2 = x^2 - 4$$

$$x^2 - 4 - x + 2 = 0$$

$$x^2 - x - 2 = 0$$

$$(x + 1)(x - 2) = 0$$

$$x + 1 = 0 \quad , \quad x - 2 = 0$$

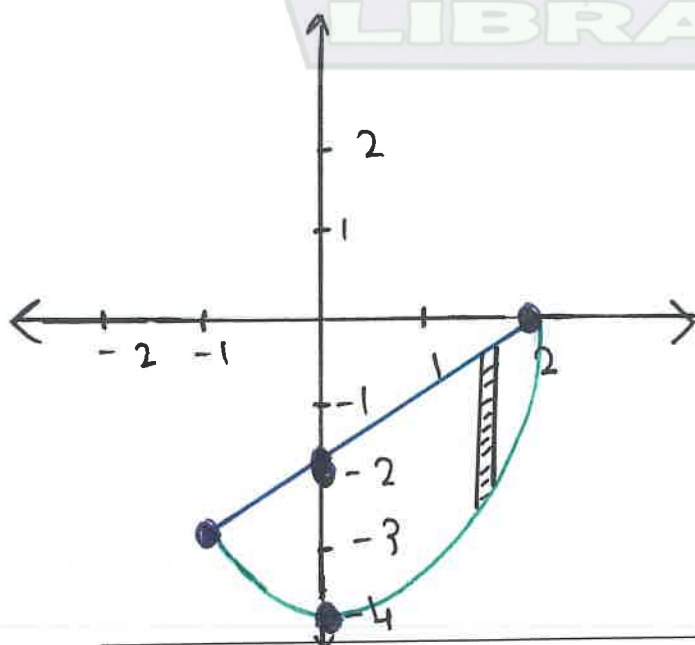
$$x = -1$$

$$x = 2$$

$$y = x - 2$$

x	-1	0	2
y	-3	-2	0

x	-1	0	2
y	-3	-4	0



$$\begin{aligned}
 A &= \int_{-1}^2 (x - 2 - (x^2 - 4)) dx \\
 &= \int_{-1}^2 (x - 2 - x^2 + 4) dx \\
 &= \int_{-1}^2 (x - x^2 + 2) dx \\
 &= \left[\frac{x^2}{2} - \frac{x^3}{3} + 2x \right]_{-1}^2 \\
 &= \left(\frac{4}{2} - \frac{8}{3} + 4 \right) - \left(\frac{1}{2} + \frac{1}{3} - 2 \right) \\
 &= 2 - \frac{8}{3} + 4 - \frac{1}{2} - \frac{1}{3} + 2 \\
 &= 8 - \frac{9}{3} - \frac{1}{2} \\
 &= 8 - 3 - \frac{1}{2} \\
 &= 5 - \frac{1}{2} = \frac{10}{2} - \frac{1}{2} = \boxed{\frac{9}{2}}
 \end{aligned}$$

Set up and sketch the area of the region bounded by the graphs of the equations

$$y = \sqrt{x-1} \quad \text{and} \quad x - y = 1$$

$$y = \sqrt{x-1}$$

$$y = x - 1$$

$$y = y$$

$$\sqrt{x-1} = x-1$$

$$(\sqrt{x-1})^2 = (x-1)^2$$

$$x-1 = x^2 - 2x + 1$$

$$x^2 - 3x + 2 = 0$$

$$(x-1)(x-2) = 0$$

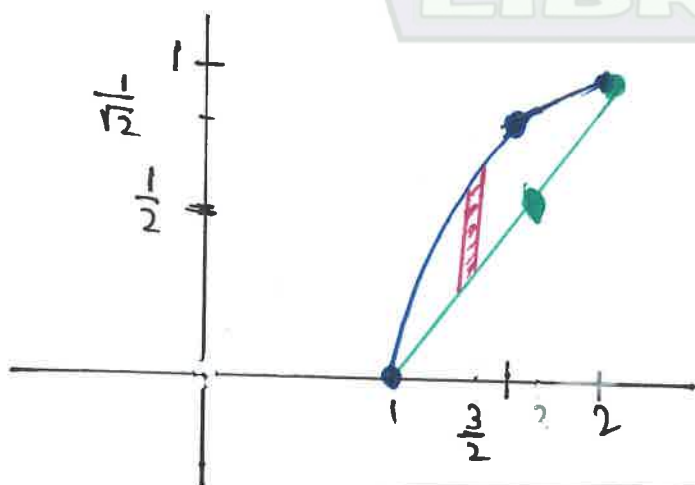
$$x=1 \quad x=2$$

$$y = \sqrt{x-1}$$

$$y = x-1$$

x	1	$\frac{3}{2}$	2
y	0	$\frac{1}{2}$	1

x	1	$\frac{3}{2}$	2
y	0	$\frac{1}{2}$	1



$$A = \int_1^2 [\sqrt{x-1} - (x-1)] dx$$

ما في
ما
Setup.

Set up^{and} sketch the area of the region bounded by the graphs of the equations

$$4x + y^2 = 12 \quad x = y$$

$$4x + y^2 = 12$$

$$4x = 12 - y^2$$

$\div 4$

$$x = 3 - \frac{1}{4}y^2$$

$$x = x$$

$$3 - \frac{1}{4}y^2 = y$$

$\times 4$

$$12 - y^2 = 4y$$

$$y^2 + 4y - 12 = 0$$

$$(y - 2)(y + 6) = 0$$

$$y = 2$$

$$y = -6$$

$$x = 3 - \frac{1}{4}y^2$$

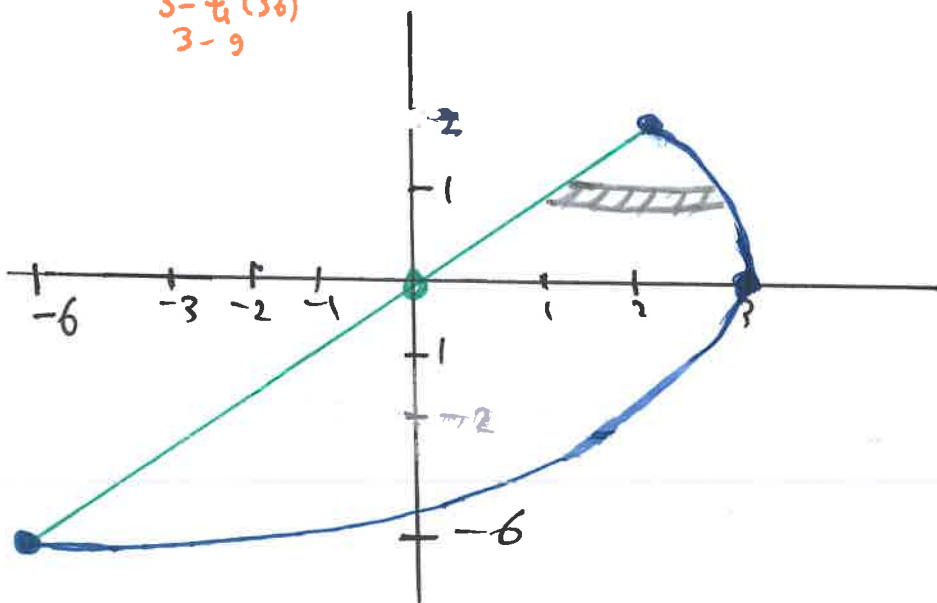
$$x = y$$

x	-6	3	2
y	-6	0	2

x	-6	0	2
y	-6	0	2

$$3 - \frac{1}{4}(36)$$

$$3 - 9$$



$$A = \int_{-6}^2 \left[3 - \frac{1}{4}y^2 - y \right] dy$$

Find the area of the region bounded by the graph of

$y = x^2$ and the lines $x = 2$ and $y = 0$

$$y = x^2$$

$$x = 2$$

$$y = 0$$

$$y = y$$

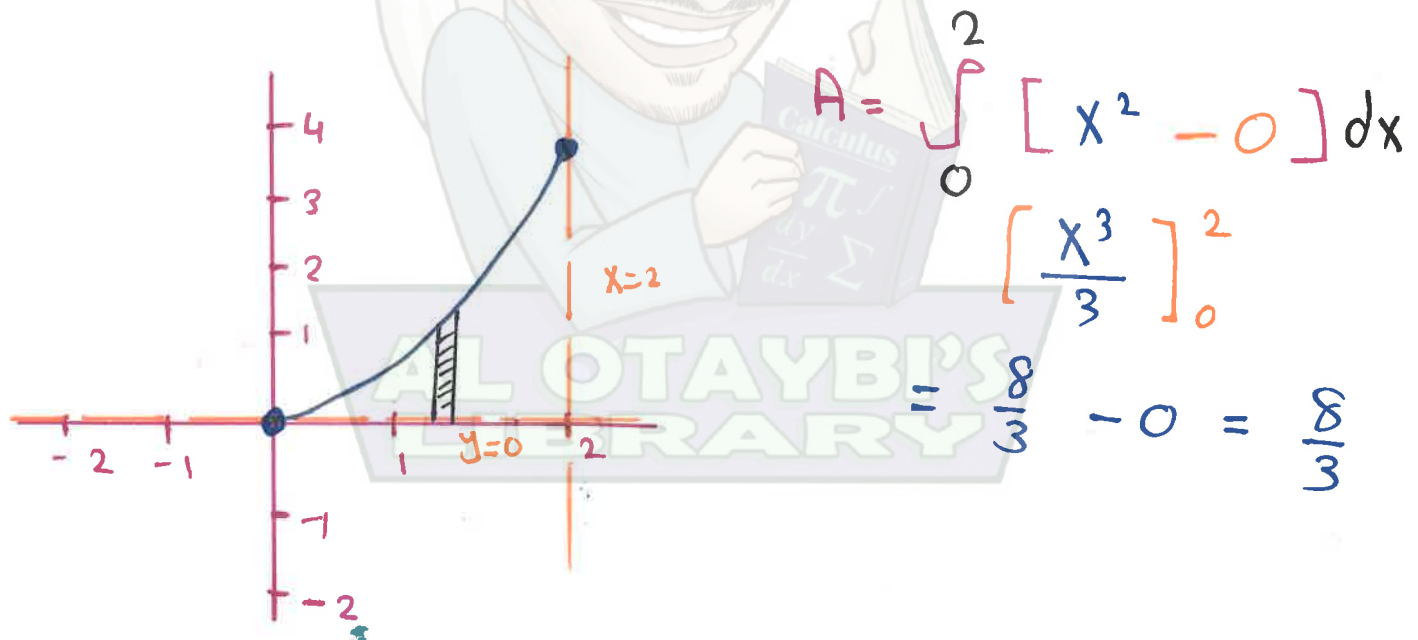
$$x^2 = 0$$

$$x = 0$$

$$y = x^2$$

x	0	2
y	0	4

حالة
التي
تعالج



يُعتبر النوع من اللامنت
يفضل الاختيار بناء على
الطرف البعيد.

a) Sketch the area of the region bounded by the graphs of the equations defined by

$C : y = \ln x$, $y = 4$, and $x = 1$

b) Set up the integral to find the area enclosed by C

$$y = \ln x$$

$$y = 4$$

$$x = 1$$

$$y = y$$

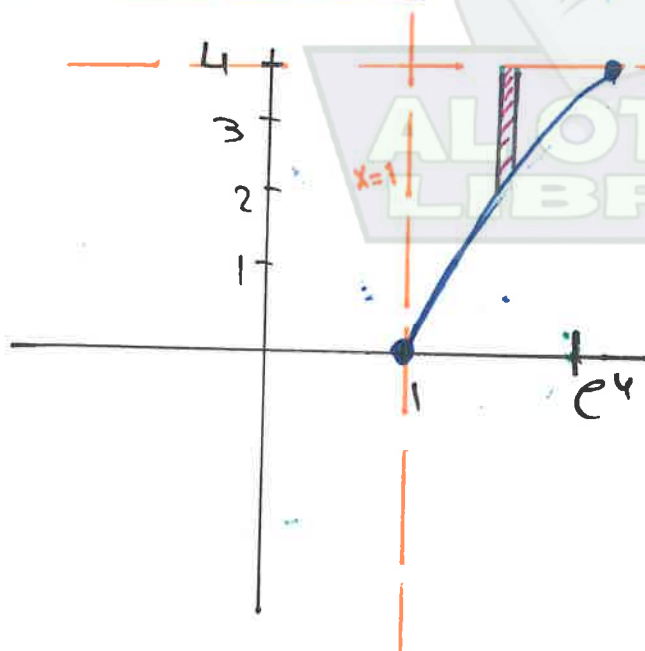
$$\ln x = 4$$

$$e^{\ln x} = e^4$$

$$x = e^4$$

$$y = \ln x$$

x	1	مال	e^4
y	0	بلي	4



$$A = \int_1^{e^4} [4 - \ln x] dx$$

ما امله
قال
Setup.

Find the area of the region bounded by the curve

$$y = x\sqrt{2x^2 + 1}, \quad y = 0, \quad x = 0 \quad \text{and} \quad x = 2$$

$$y = x\sqrt{2x^2 + 1}$$

$$y = y$$

$$x\sqrt{2x^2 + 1} = 0$$

$$x = 0 \quad \sqrt{2x^2 + 1} = 0$$

$$2x^2 + 1 = 0$$

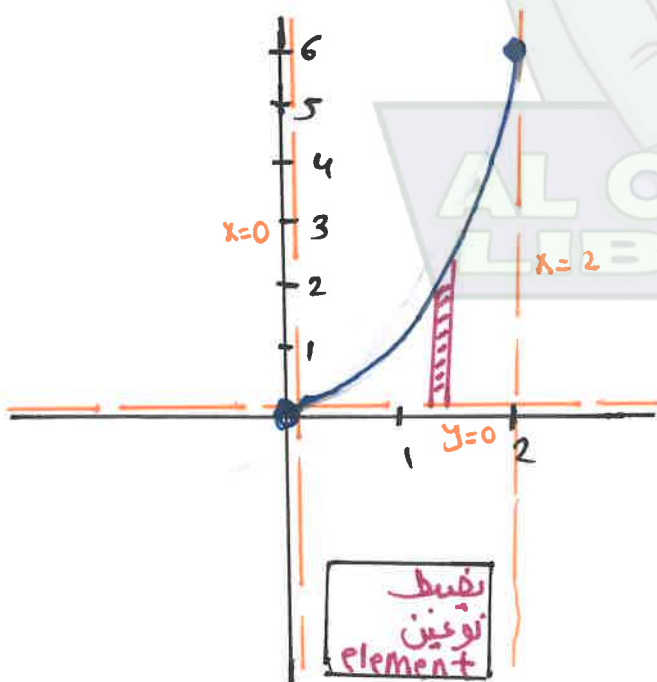
$$2x^2 = -1$$

None

$$y = x\sqrt{2x^2 + 1}$$

x	0	عالية	2
y	0	دنيا	6

2√9



$$A = \int_0^2 [x\sqrt{2x^2 + 1} - 0] dx$$

$$= \int_0^2 x(2x^2 + 1)^{\frac{1}{2}} dx$$

$$u = 2x^2 + 1$$

$$du = 4x dx \rightarrow \frac{1}{4} du = x dx$$

$$= \frac{1}{4} \int u^{\frac{1}{2}} du = \frac{1}{4} \cdot \frac{2}{3} u^{\frac{3}{2}}$$

$$= \frac{1}{6} [(2x^2 + 1)^{\frac{3}{2}}]_0^2$$

$$= \frac{1}{6} [9^{\frac{3}{2}} - 1^{\frac{3}{2}}]$$

$$= \frac{1}{6} [(3^2)^{\frac{3}{2}} - 1]$$

$$= \frac{1}{6} [27 - 1] = \frac{1}{6} \cdot 26 = \frac{13}{3}$$

Find the area of the region bounded by the graph of the equation

$x + y^2 = 4$ and the line $x = 0$

$x = 4 - y^2$

$x = 0$

$x = x$

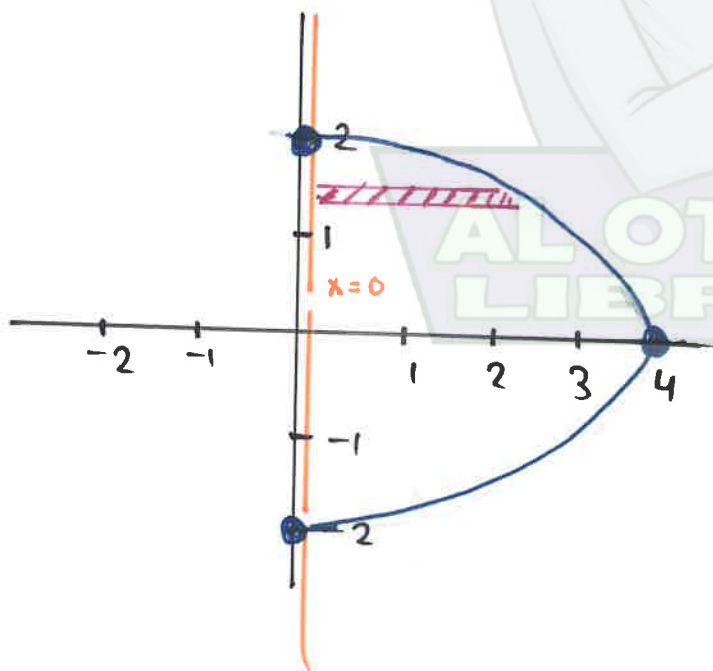
$4 - y^2 = 0$

$(2 - y)(2 + y) = 0$

$y = 2$ $y = -2$

$x = 4 - y^2$

x	0	4	0
y	-2	0	2



$A = \int_{-2}^2 [4 - y^2 - 0] dy$

$\left[4y - \frac{y^3}{3} \right]_{-2}^2$

$(8 - \frac{8}{3}) - (-8 + \frac{8}{3})$

$8 - \frac{8}{3} + 8 - \frac{8}{3}$

$16 - \frac{16}{3}$

$\frac{48}{3} - \frac{16}{3} = \frac{32}{3}$

Find the area of the region enclosed by the curves

$$y = 4 - x^2 \text{ and } y = 3$$

Jan 2017

$$y = 4 - x^2$$

$$y = 3$$

$$y = y$$

$$4 - x^2 = 3$$

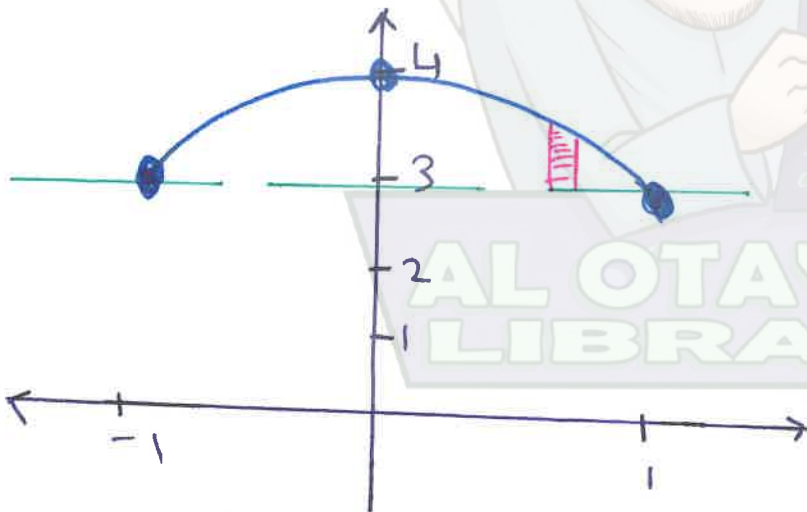
$$1 = x^2$$

$$x = -1$$

$$x = 1$$

$$y = 4 - x^2$$

x	-1	0	1
y	3	4	3



$$A = \int_{-1}^1 (4 - x^2 - 3) dx$$

$$= \int_{-1}^1 (1 - x^2) dx$$

$$= \left[x - \frac{x^3}{3} \right]_{-1}^1 = \left(1 - \frac{1}{3} \right) - \left(-1 + \frac{1}{3} \right) = \frac{2}{3} - \left(-\frac{2}{3} \right) = \left(\frac{4}{3} \right)$$

Find and sketch the area of the region bounded by the graphs of the equations

$y = e^x$, $y = x^2 - 1$, from $x = -1$ to $x = 1$

$y = e^x$

$y = x^2 - 1$

$x = -1$

$x = 1$

$y = y$

$e^x = x^2 - 1$

$e^x - x^2 = -1$

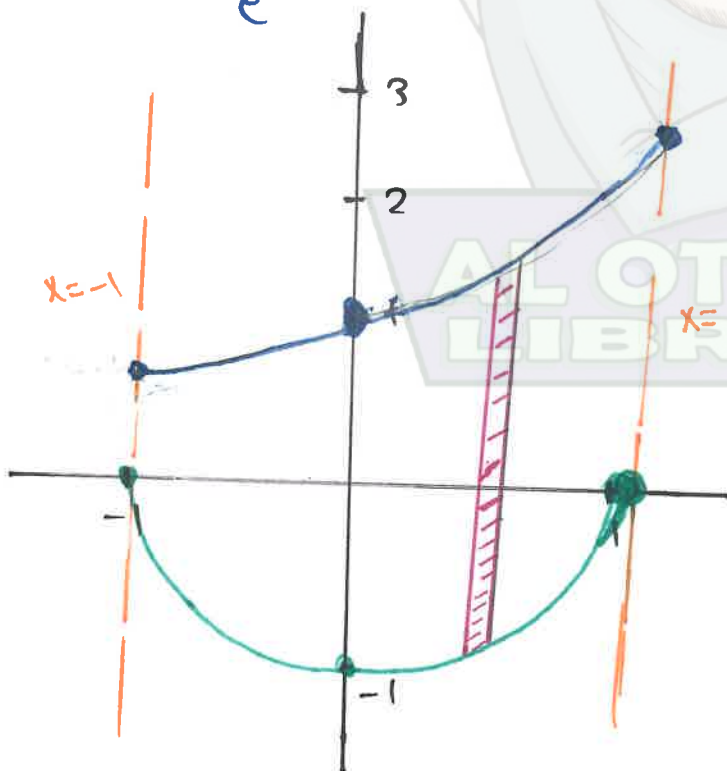
ما أفعل
طافا

$y = e^x$

x	-1	0	1
y	e^{-1}	1	e^1

$y = x^2 - 1$

x	-1	0	1
y	0	-1	0



$A = \int_{-1}^1 [e^x - (x^2 - 1)] dx$

$\left[e^x - \frac{x^3}{3} + x \right]_{-1}^1$

$(e^1 - \frac{1}{3} + 1) - (e^{-1} + \frac{1}{3} - 1)$

$e^1 - \frac{1}{3} + 1 - e^{-1} - \frac{1}{3} + 1$

$e^1 - e^{-1} - \frac{2}{3} + 2$

Set up an integral that can be used to find the area of the region bounded by the x -axis and the curve of $y = x^3 - 9x$

$$y = x^3 - 9x$$

$$y = 0$$

$$y = y$$

$$x^3 - 9x = 0$$

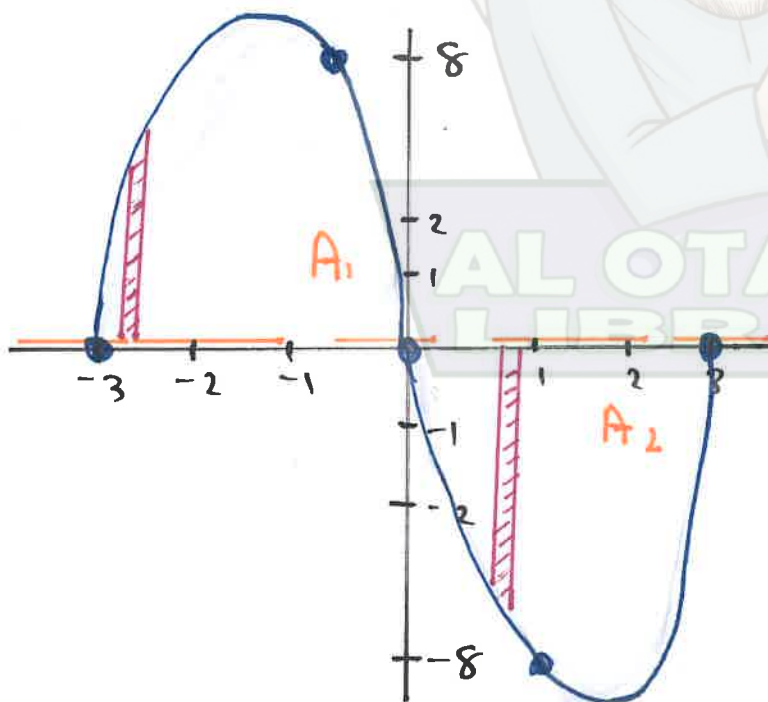
$$x(x^2 - 9) = 0$$

$$x(x - 3)(x + 3) = 0$$

$$x = 0 \quad x = 3 \quad x = -3$$

$$y = x^3 - 9x$$

x	-3	-1	0	1	3
y	0	8	0	-8	0



$$A_1 = \int_{-3}^0 [x^3 - 9x - 0] dx$$

$$A_2 = \int_0^3 [0 - (x^3 - 9x)] dx$$

$$\Rightarrow A_T = A_1 + A_2$$

Solution #2
By Symmetry
 $A_T = 2A_1$

Find and sketch the area of the region bounded by the graphs of the equations

$y = \sqrt{x} - 2$, x - axis , from $x = 0$ to $x = 9$

$$y = \sqrt{x} - 2$$

$$y = y$$

$$\sqrt{x} - 2 = 0$$

$$\sqrt{x} = 2$$

$$x = 4$$

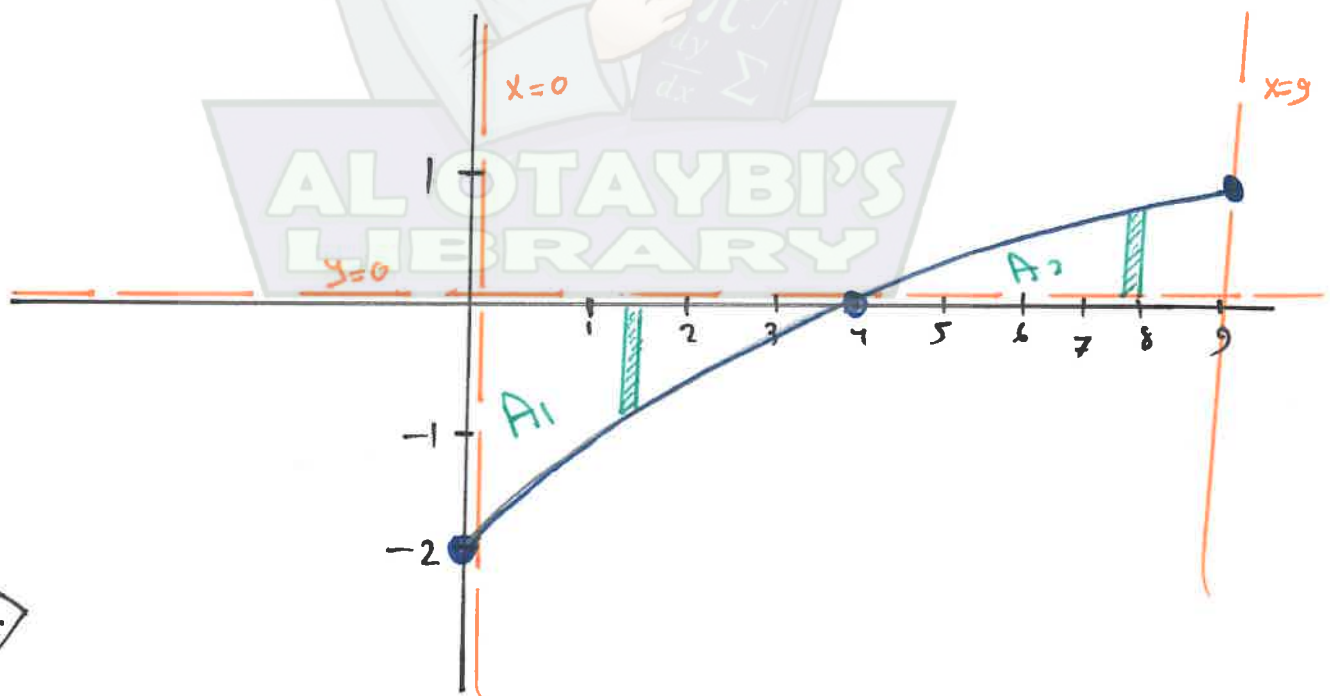
$$y = \sqrt{x} - 2$$

x	0	4	9
y	-2	0	1

$$y = 0$$

$$x = 0$$

$$x = 9$$



$$\begin{aligned}
 A_1 &= \int_0^4 [0 - (\sqrt{x} - 2)] dx \\
 &= \int_0^4 [-x^{\frac{1}{2}} + 2] dx \\
 &= \left[-\frac{x^{\frac{3}{2}}}{\frac{3}{2}} + 2x \right]_0^4 = \left[-\frac{2}{3} (4)^{\frac{3}{2}} + 8 - 0 \right] \\
 &= -\frac{2}{3} 8 + 8 = -\frac{16}{3} + \frac{24}{3} = \frac{8}{3} \quad \text{with note } (2^2)^{\frac{3}{2}}
 \end{aligned}$$

$$\begin{aligned}
 A_2 &= \int_4^9 [\sqrt{x} - 2 - 0] dx \\
 &= \int_4^9 [x^{\frac{1}{2}} - 2] dx \\
 &= \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} - 2x \right]_4^9 = \left[\frac{2}{3} (9)^{\frac{3}{2}} - 18 \right] - \left[\frac{2}{3} (4)^{\frac{3}{2}} - 8 \right] \\
 &= \frac{2}{3} (27) - 18 - \left[\frac{2}{3} 8 - 8 \right] \quad \text{with notes } (3^2)^{\frac{3}{2}} \text{ and } (2^2)^{\frac{3}{2}} \\
 &= 18 - 18 - \left[\frac{16}{3} - \frac{24}{3} \right] \\
 &= - \left[-\frac{8}{3} \right] = \frac{8}{3}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow A_T &= A_1 + A_2 \\
 &= \frac{8}{3} + \frac{8}{3} = \frac{16}{3}
 \end{aligned}$$

Find the area of the region bounded by the graphs of the equations :

$$y = x \quad \text{and} \quad y = x^3$$

$$y = x$$

$$y = x^3$$

$$y = y$$

$$x = x^3$$

$$x - x^3 = 0$$

$$x(1 - x^2) = 0$$

$$x(1 - x)(1 + x) = 0$$

$$x = 0 \quad x = 1 \quad x = -1$$

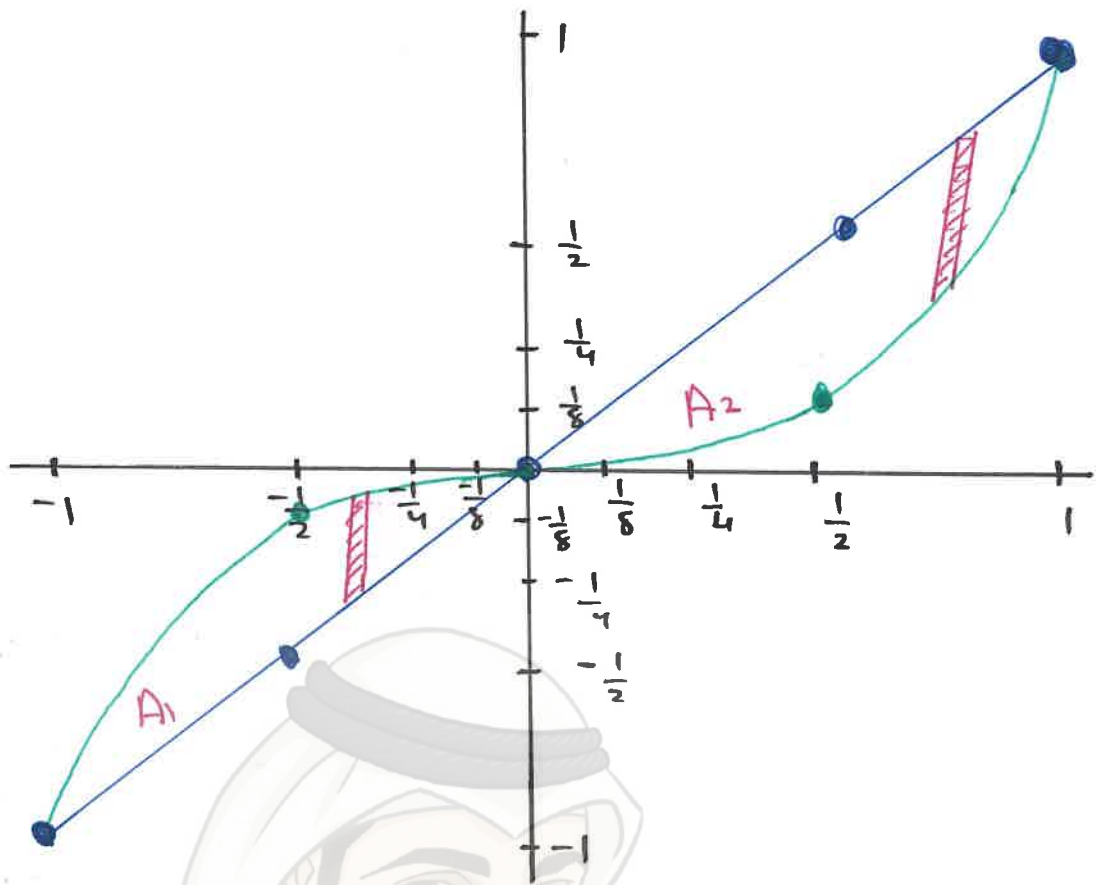
$$y = x$$

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
y	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

$$y = x^3$$

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
y	-1	$-\frac{1}{8}$	0	$\frac{1}{8}$	1

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$$\begin{aligned}
 A_1 &= \int_{-1}^0 [x^3 - x] dx \\
 &= \left(\frac{x^4}{4} - \frac{x^2}{2} \right)_{-1}^0 \\
 &= 0 - \left(\frac{1}{4} - \frac{1}{2} \right) \\
 &= - \left(\frac{1}{4} - \frac{2}{4} \right) \\
 &= - \left(-\frac{1}{4} \right) = \frac{1}{4}
 \end{aligned}
 \quad \Bigg| \quad
 \begin{aligned}
 A_2 &= \int_0^1 [x - x^3] dx \\
 &= \left(\frac{x^2}{2} - \frac{x^4}{4} \right)_0^1 \\
 &= \left(\frac{1}{2} - \frac{1}{4} \right) - 0 \\
 &= \frac{2}{4} - \frac{1}{4} = \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow A_T &= A_1 + A_2 \\
 &= \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}
 \end{aligned}$$

Let R be the region between the curves $y = x$ and $y = x^2$ from $x = 0$ to $x = 2$.

Sketch the region of R , and find its area.

May 2017

$$y = x$$

$$y = x^2$$

$$x = 0$$

$$x = 2$$

$$y = y$$

$$x = x^2$$

$$x - x^2 = 0$$

$$x(1 - x) = 0$$

$$x = 0$$

$$x = 1$$

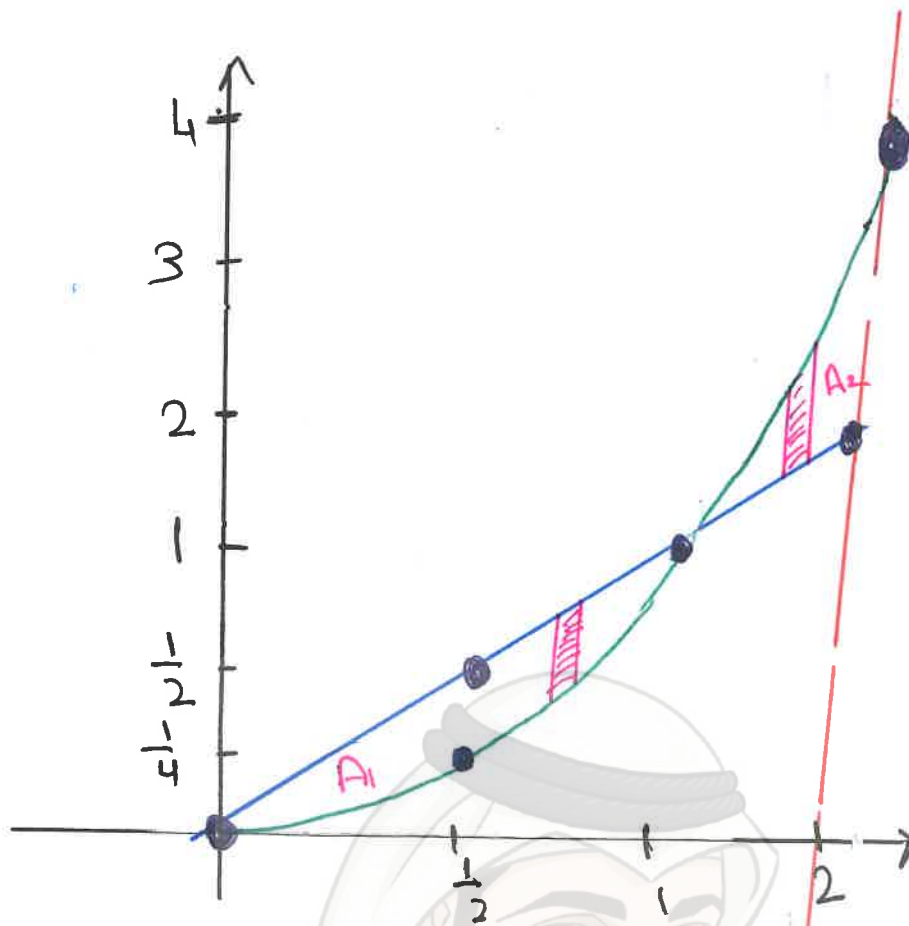
$$y = x$$

x	0	$\frac{1}{2}$	1	2
y	0	$\frac{1}{2}$	1	2

$$y = x^2$$

x	0	$\frac{1}{2}$	1	2
y	0	$\frac{1}{4}$	1	4

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$$A_1 = \int_0^1 (x - x^2) dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1$$

$$= \frac{1}{2} - \frac{1}{3} - 0$$

$$A_2 = \int_1^2 (x^2 - x) dx$$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2$$

$$= \left(\frac{8}{3} - \frac{4}{2} \right) - \left(\frac{1}{3} - \frac{1}{2} \right)$$

$$= \frac{8}{3} - \frac{4}{2} - \frac{1}{3} + \frac{1}{2}$$

$$= \frac{7}{3} - \frac{3}{2}$$

$$A_T = A_1 + A_2$$

$$= \frac{1}{2} - \frac{1}{3} + \frac{7}{3} - \frac{3}{2}$$

$$= \frac{6}{3} - \frac{2}{2}$$

$$= 2 - 1 = 1$$

Let R be the region between the curves $y = x^2$ and $y = \sqrt{x}$ from $x = 0$ to $x = 4$.

Sketch the region of R , and find its area.

Aug 2017

$$y = x^2$$

$$y = \sqrt{x}$$

$$x = 0$$

$$x = 4$$

$$y = y$$

$$x^2 = \sqrt{x}$$

$$x^4 = x$$

$$x^4 - x = 0$$

$$x(x^3 - 1) = 0$$

$$x = 0$$

$$x^3 - 1 = 0$$

$$x^3 = 1$$

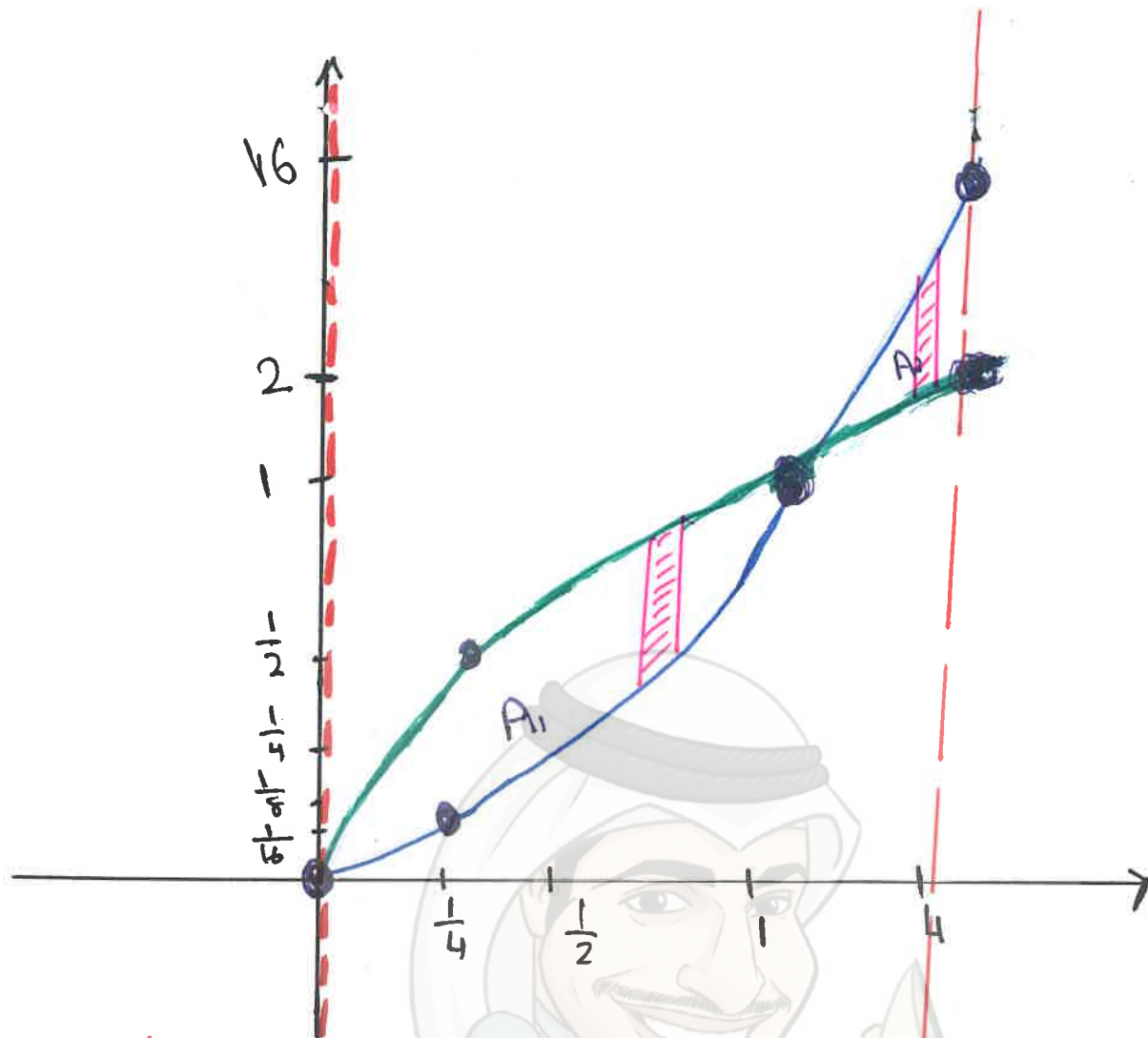
$$x = 1$$

$$y = x^2$$

$$y = \sqrt{x}$$

x	0	$\frac{1}{4}$	1	4
y	0	$\frac{1}{16}$	1	16

x	0	$\frac{1}{4}$	1	4
y	0	$\frac{1}{2}$	1	2



$$A_1 = \int_0^1 (\sqrt{x} - x^2) dx$$

$$= \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{x^3}{3} \right]_0^1$$

$$= \frac{2}{3} - \frac{1}{3} - 0$$

$$= \frac{1}{3}$$

$$A_2 = \int_1^4 (x^2 - \sqrt{x}) dx$$

$$= \left[\frac{x^3}{3} - \frac{2}{3} x^{\frac{3}{2}} \right]_1^4$$

$$= \left(\frac{64}{3} - \frac{2}{3}(8) \right) - \left(\frac{1}{3} - \frac{2}{3} \right)$$

$$= \frac{64}{3} - \frac{16}{3} - \frac{1}{3} + \frac{2}{3}$$

$$= \frac{49}{3}$$

$$\Rightarrow A_T = A_1 + A_2$$

$$= \frac{1}{3} + \frac{49}{3}$$

$$= \frac{50}{3}$$



Find and sketch the area of the region bounded by the graphs of the equations

$$y = |x| \quad \text{and} \quad y = x^2 - 2 \quad \text{and} \quad x \geq 0$$

$$y = |x|$$

$$\swarrow \searrow$$

$$y = x \quad y = -x$$

مرفوضه
لأن قال
x > 0

$$y = x^2 - 2$$

$$[0, \infty)$$

$$y = x$$

$$y = x^2 - 2$$

$$y = y$$

$$x = x^2 - 2$$

$$x^2 - x - 2 = 0$$

$$(x + 1)(x - 2) = 0$$

$$x = -1$$

$$x = 2$$

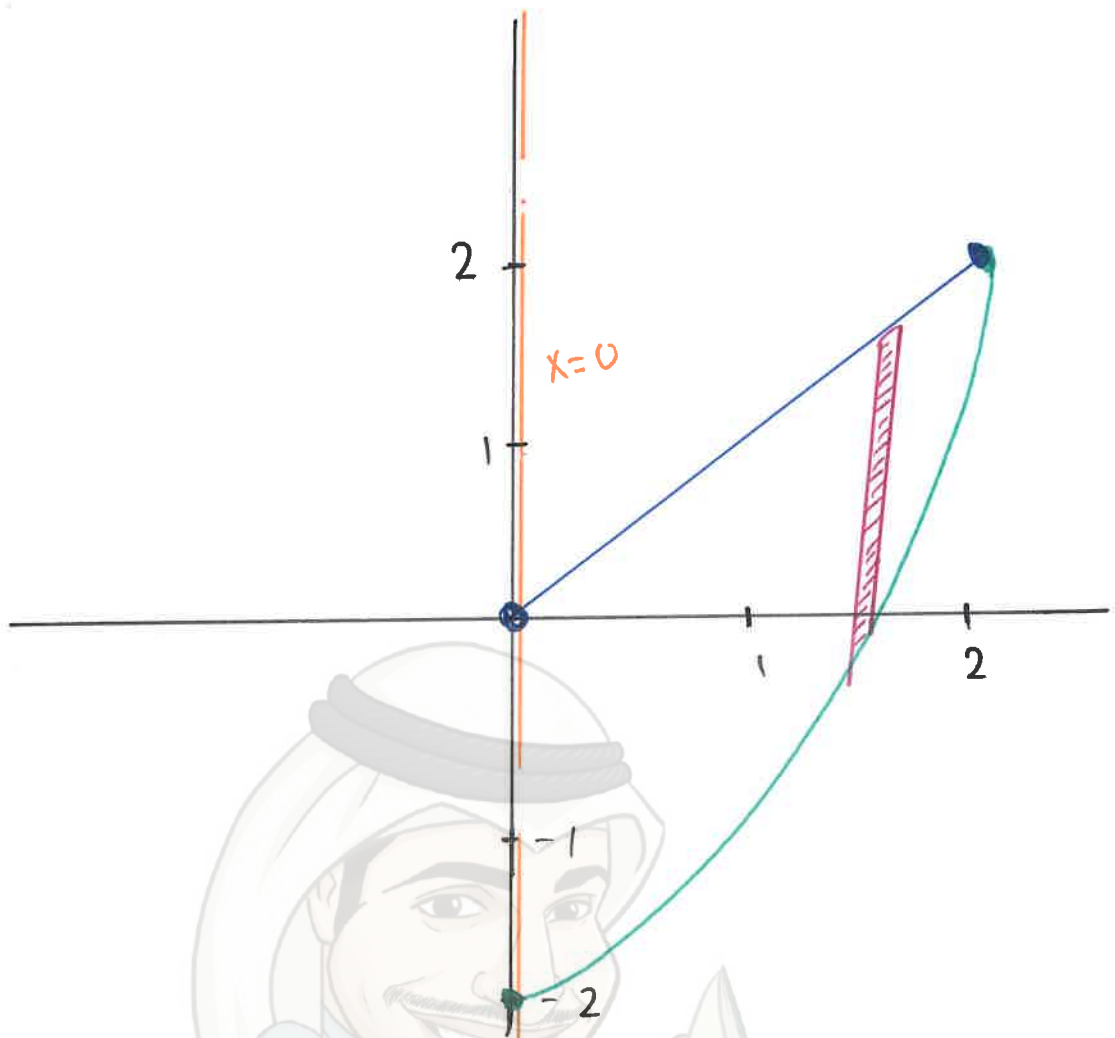
مرفوضه
rejected

$$y = x$$

x	0	2
y	0	2

$$y = x^2 - 2$$

x	0	2
y	-2	2



$$\begin{aligned}
 A &= \int_0^2 (x - (x^2 - 2)) dx \\
 &= \left(\frac{x^2}{2} - \frac{x^3}{3} + 2x \right) \Big|_0^2 \\
 &= \frac{4}{2} - \frac{8}{3} + 4 - 0 \\
 &= 2 - \frac{8}{3} + 4 \\
 &= 6 - \frac{8}{3} + \\
 &= \frac{18}{3} - \frac{8}{3} = \frac{10}{3}
 \end{aligned}$$

Find the area of the region bounded by the graphs of the equations

$$x^3 - y = 0, \quad x + y = 0 \quad \text{and} \quad y = 8$$

$$y = x^3$$

$$y = -x$$

$$y = 8$$

$$y = y$$

$$x^3 = -x$$

$$x^3 + x = 0$$

$$x(x^2 + 1) = 0$$

$$x = 0$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

No one

$$y = x^3 \rightarrow x = y^{\frac{1}{3}}$$

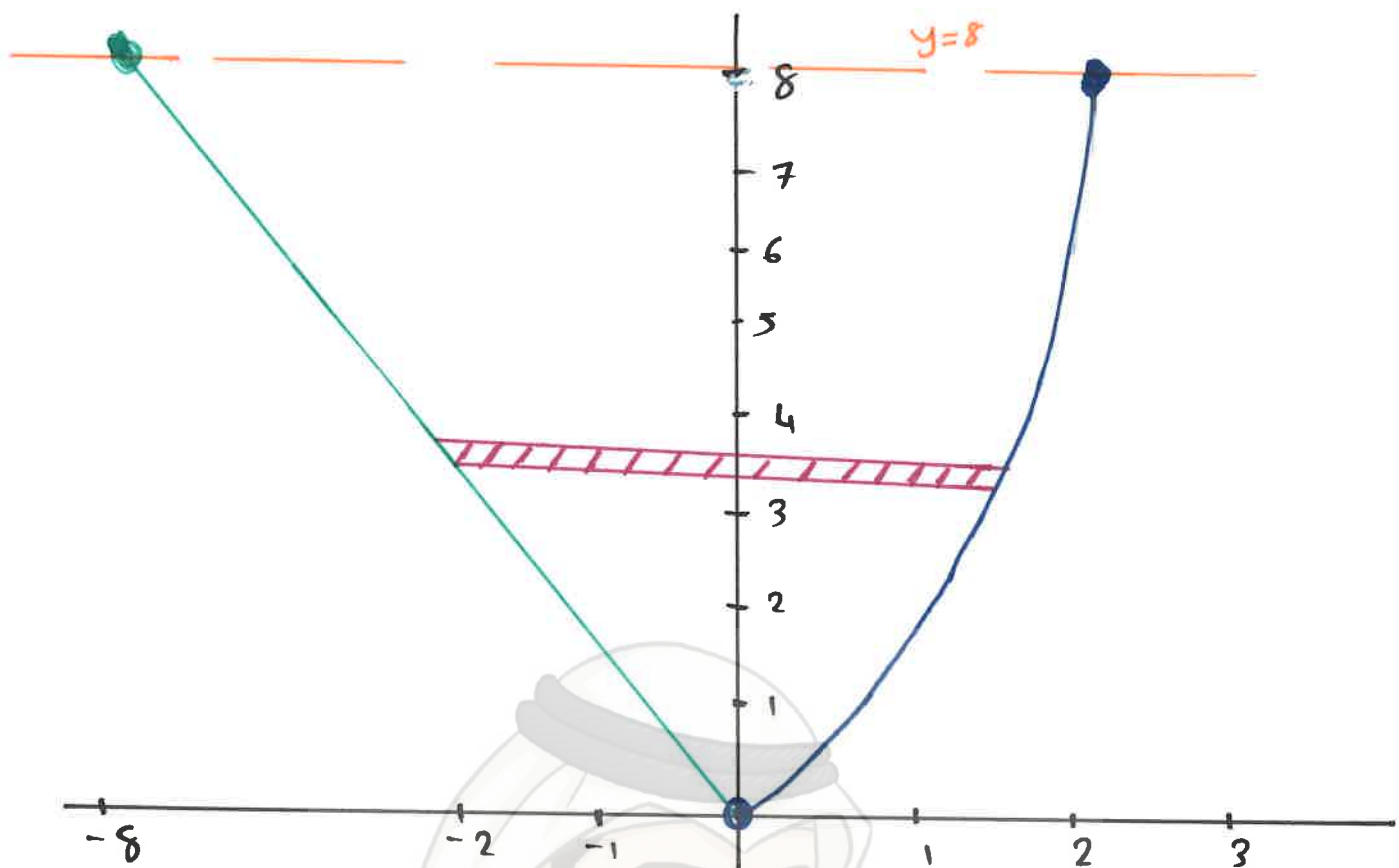
x	0	قال	2
y	0	قال	8

$$y = -x \rightarrow x = -y$$

x	0	قال	-8
y	0	قال	8

← قال داي اخذ رقم بينهم
لكن لو جيتي لازم تاخذين
بين 2 لانهم مختلفين





$$A = \int_0^8 [x^3 - (-x)] dy$$

$$\begin{aligned}
 A &= \int_0^8 [y^{\frac{1}{3}} - (-y)] dy \\
 &= \left[\frac{y^{\frac{4}{3}}}{\frac{4}{3}} + \frac{y^2}{2} \right]_0^8 = \left[\frac{3}{4} (8)^{\frac{4}{3}} + \frac{64}{2} - 0 \right] \\
 &= \frac{3}{4} 16 + 32 \\
 &= 3(4) + 32 \\
 &= 12 + 32 = 44
 \end{aligned}$$

لازم نلای بعض

AL OTAYBI'S LIBRARY

(2^3)^{4/3}